

Sovereign Depths: An Underlake Data Center Framework for Intelligence Infrastructure and Indonesian National Resilience

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September 2026

Abstract

Indonesian strategic data currently rests either in terrestrial data centers concentrated in a few urban corridors or in foreign clouds subject to foreign legal process. This paper asks a question the underwater data center literature has not isolated: under what conditions, if any, does submerging a modular data center in an inland freshwater lake offer capability that a matched domestic terrestrial facility, sited at the same lake with the same energy access and the same multi-site redundancy, does not? We develop an assessment framework that separates the five candidate benefits of the concept, domestic jurisdiction, remote siting, energy proximity, multi-site replication, and submersion itself, and tests them one at a time. The framework contains: a bounded threat model with adversary objectives and a published ordinal score matrix; an explicit service definition (two 50 kW modules, synchronous replication with witness-based fencing, and both a two-of-four capacity criterion and a three-of-four majority-quorum criterion); a screening of seven Indonesian lakes in which five pass a published 100 m depth filter and are ranked with published scores and one-at-a-time weight sensitivity; screening-level thermal balances (a basin-average anomaly of 1.0×10^{-3} K over fifty years and a line-plume dilution to 9.9×10^{-4} K at the thermocline base, with the twelve-month footprint expressed as a heat-equivalent radius of 228–509 m under prescribed detection thresholds); a ten-year cost model at equal service whose decisive comparator, a hardened terrestrial facility with lake-water cooling, costs 6.34 against 7.85 million dollars for the submerged design and 6.00 for conventional cooling; and a fully documented Monte Carlo model of correlated failure in a four-site network, which yields a two-of-four capacity

availability of 0.9952 and a three-of-four quorum availability of 0.9484, with recovery after a regional seismic event returning to the long-run level within four to six months. The conclusion is deliberately conditional: inland submersion removes selected direct maritime access paths and offers plausible physical-access trade-offs, but on the evidence assembled here it does not yet demonstrate superiority over equally distributed domestic terrestrial alternatives. The paper ends with the gated measurement program under which the hypothesis could be confirmed or rejected.

Keywords: underwater data center; freshwater lake data center; data sovereignty; national resilience; threat model; multi-criteria analysis; sensitivity analysis; Monte Carlo simulation; technology innovation system; Indonesia.

1 Introduction

Indonesia’s need for sovereign computing is a premise of national policy rather than a finding of this paper; the open question is where such computing should physically sit.

The legal baseline must be stated precisely, because it is often overstated. Under the electronic system operator regulation, PP 71/2019, public sector electronic system operators are obliged to place their systems and data centers within Indonesian territory, subject to statutory exceptions where the required storage technology is not available domestically (Pasal 20 ayat (2)–(5)); private sector operators may operate domestically and/or abroad under supervision and enforcement conditions, with further sectoral rules for financial services (Pasal 21) [32]. The personal data protection law, UU 27/2022, governs cross-border transfer of personal data through an ordered set of conditions: a protection level at least equivalent to Indonesian law, failing that adequate and binding protection safeguards, and failing that the data subject’s consent (Pasal 56) [53]. The state intelligence law, UU 17/2011,

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adds a stricter layer for intelligence data: secrecy and compartmentation are statutory principles of intelligence activity (Pasal 2 huruf b–c), intelligence personnel must keep secret the special facilities and equipment of intelligence activity (Pasal 18 huruf b), intelligence secrets carry a protection and storage period (Masa Retensi) of 25 years extendable with the approval of the House of Representatives (Pasal 1 angka 7; Pasal 25 ayat (3)–(4)), and intentional or negligent disclosure carries criminal penalties (Pasal 44–45) [52]. None of these statutes mandates a lake. What they establish is that strategic state data belongs under domestic control; the physical form of that control remains an engineering and policy choice among domestic options, and the retention duty concerns information and its storage, not the physical lifetime of any single storage medium.

The existing arrangements are unsatisfactory in specific, testable ways, not in all possible ways. Terrestrial data centers in Indonesia are reported to be concentrated around the Jakarta and Batam corridors; we treat this as an industry observation that a systematic survey should confirm, and we use the resilience literature only for the general proposition that geographic concentration creates correlated risk [22]. Foreign cloud platforms, the second arrangement, can carry data beyond the effective reach of Indonesian courts, although the strength of that reach depends on contractual terms, key management, and bilateral cooperation rather than on location alone; the data sovereignty literature documents the resulting loss of control in detail [13, 39]. Submarine cables, the connective tissue of the digital economy, are themselves a recognized vulnerability surface [8, 38].

This paper examines a third option: the freshwater lake data center (FLDC), a sealed modular data center deployed at depth in an inland lake, connected to a hardened shore base by a power and fiber umbilical. Underwater data centers exist at sea, and the engineering literature on them is real but small [31, 42, 50]. The concept of submerging modules in bodies of water other than the ocean, including lakebeds, was claimed in the Microsoft patent [41], and server-scale submerged hardware has been tested in lakes by the AquaSonic group [43]. Within the scope of a structured literature search (OpenAlex and Crossref by DOI, OpenAlex title/abstract queries for underwater and submerged data centers, lake thermal pollution, and critical infrastructure resilience, restricted to open-access or metadata-verifiable records, cut-off September 2026), we found no peer-reviewed threat model, thermal model, or assessment framework for inland submerged data centers, and no production facility. The contribution claimed here is that framework, in exactly that scope; the claims of prior-art absence are bounded by that search protocol, not by an assertion about all conceivable literature.

The central claim is that a lake is not a small ocean. Indonesian lakes are internal waters, and installations at their bottom sit unambiguously within Indonesian territory, which removes international contestation over ju-

risdiction, though not domestic permitting, customary rights, or supply-chain questions. The threat environment is defined by land, air, internal adversaries, and a sound channel through the water itself, rather than by foreign navies. This shift changes the design problem in ways that may favor the defender, but the favor must be demonstrated rather than asserted. We therefore separate, throughout the paper, the five bundled benefits of the lake concept: (i) domestic jurisdiction, (ii) remote inland siting, (iii) proximity to hydropower, (iv) multi-site redundancy across regions, and (v) submersion itself. Benefits (i)–(iv) are available to domestic terrestrial facilities as well; the research question of Section 7 is what submersion adds after the other four are held equal. Every number in this paper carries one of four statuses, known, assumed, computed, or unproven, stated where it appears and summarized in Section 12; the deterministic calculations and the simulation are provided as open companion code at <https://github.com/jalaranu/companion> so that every result can be recomputed.

The paper proceeds as follows. Section 2 positions the work against the underwater data center and resilience literatures. Section 3 develops the inland threat model and the adversary objectives. Section 4 presents the system design and the service definition. Section 5 screens candidate Indonesian lakes, publishes the ranking inputs, and reports the sensitivity analysis. Section 6 reports the screening-level thermal calculations. Section 7 compares the lake option against hardened terrestrial, lake-cooled terrestrial, and bunker facilities at equal service. Section 8 develops the Lake Matano case study. Section 9 models common-mode failure and simulated recovery. Section 10 maps military and intelligence requirements and the interdisciplinary frame. Section 11 analyzes the innovation system and sets out a gated roadmap. Section 12 states the evidence status and the honest limits, and Section 13 concludes.

2 Related Work and the Gap

Three literatures bear on the problem. The first is engineering work on underwater data centers. The reference deployment is Microsoft’s Project Natick, whose intellectual foundation is the submerged data center patent of [41], claiming modular deployment, environmental protection by submersion, and marine renewable energy integration, for bodies of water including lakes. Published follow-on work has concentrated on narrow problems: resilience to marine heat waves and reservoir-based cooling [31], sub-sea power distribution [35], and the water footprint of data centers generally [26]. One security result matters disproportionately for this paper. Sheldon and colleagues demonstrated that acoustic signals injected through the water column can degrade RAID throughput and remove filesystem nodes in submerged hardware, first against hard disk storage [42] and then against a hybrid SSD–HDD

distributed system, with open-water trials at ranges up to 6.35 m limited by the test facility [43]. These are testbed experiments against rotating-media and hybrid configurations, not validations of an all-flash facility, and no part of this paper should be read as claiming that a surface attacker can degrade a module at 300 m; that transfer is an open question that the pilot of Section 11 must test with the storage bill of materials the design actually uses. Critical social analysis of subsea data centers has begun to appear [50]; it asks whose futures these machines serve, a question this paper takes seriously through its governance analysis rather than answering.

The second literature concerns lake physics and lake ecology, and it is decisive for design. Thermal pollution measurably alters stratification and circulation in lakes [56], and climate change is already altering lake mixing regimes worldwide [59] while temperate lakes lose oxygen [15]. Deep tropical lakes such as Matano are persistently stratified, and Katsev and colleagues measured how mixing varies with depth and how transport governs oxygen, iron, methane, and phosphorus cycling; the useful design datum is that profile and its uncertainty, not the label “stratified” alone [17, 47]. Meromictic lakes, in which deep waters never overturn [27], still sustain local mixing and transport processes, so a no-overturn label does not license treating the deep layer as a sealed reservoir. These results establish the central ecological constraint of this paper: a lake data center must not reject heat into the water column at a rate that measurably destabilizes stratification, and the burden of demonstrating compliance sits on the proponent.

The third literature is on data sovereignty and critical infrastructure resilience. The sovereignty literature distinguishes control of data from its physical location and documents how infrastructure choices quietly settle both [13, 16, 39]. The resilience literature supplies assessment frameworks built around terrestrial grids, pipelines, and bridges [4, 61, 58]. Within the search scope stated above, neither literature contains a threat model for inland submerged infrastructure, a screening thermal model for submerged heat rejection in a stratified tropical lake, or a cost comparison at equal service against terrestrial alternatives. Those are the gaps this paper fills.

3 Threat Model: Inland Waters and the National Security Calculus

A threat model states which adversary can do what to which asset, at what cost. For undersea infrastructure the adversaries are maritime: submarines that tap or cut cables, surface vessels that drag anchors, and states that assert contested jurisdiction. The legal layer is equally maritime. The high seas freedom to lay cables coexists with coastal state authority in ways that generate genuine ambiguity [5], and major powers interpret the relevant

law differently [21]. Cable security accordingly consumes a dedicated sensor and monitoring literature [6].

Only some of this transfers to a lake. A deep lake in Sulawesi or Sumatra cannot be entered by a foreign submarine; candidate lakes lie tens to hundreds of kilometres from any open-sea approach, with no navigable connection to the sea (distances must be verified per site rather than assumed). Candidates are not on maritime shipping lanes, although local boat traffic remains. Their legal status as internal waters removes international contestation over the installation; what remains is the full domestic stack of permits, customary rights, and supply-chain dependencies. In short, the maritime threat paths narrow substantially; the land, air, insider, cyber, and shore-side paths remain, and one new channel, acoustic transmission through the water column, is added.

What remains is a different adversary set, listed explicitly. First, overland attack: a land-based adversary, state or non-state, must approach through Indonesian territory, through population that can observe, and past a shore base that can be defended; whether this observability actually deters a determined adversary is a scenario question, not a certainty. Second, insider threat: the human factor is comparable across deployment classes and is managed by personnel security together with architectural controls, separation of duties, dual approval, key management, segmentation, and audit logging. Third, standoff attack from the air: drones and standoff munitions can reach inland targets; a module at depth reduces direct optical aimability, and whether water attenuates blast and fragmentation efficiently for the munitions of concern requires scenario-specific fluid–structure analysis, not assertion. Fourth, seismic hazard: Indonesian deep lakes sit on or near active faults, and seismic exposure at the candidate sites is plausibly the one category in which inland deployment is worse than the terrestrial comparison; site-specific values must come from the national hazard map, not from a different earthquake elsewhere. Fifth, domestic unrest: Indonesia’s history of mass demonstrations is real, and the design response is remoteness and redundancy; fortification offers limited value against large-scale coordinated unrest, and peaceful protest must not be conflated with sabotage. Sixth, acoustic attack, a category identified by the underwater data center literature itself. Sound propagates far in water, and a malicious acoustic signal can drive disk resonance and degrade storage throughput, demonstrated empirically against rotating-media and hybrid testbeds [42, 43]. The attacker needs a surface craft and a transducer rather than access to the module, but range, power, frequency, and the target’s storage technology all condition effectiveness, none of which is established for the proposed configuration. An inland lake makes such an attack easier to localize and attribute than in the open sea, and the design must answer it with hydrophones, vibration isolation specified by measured transfer functions, and anomaly detection rather than with complacency. Figure 1 summarizes the shift; the

underlying scores are published in Table 1.

The comparison can be stated formally. For a deployment class c , the aggregate risk is

$$R_c = \sum_{t \in T} L_{t,c} I_{t,c}, \quad (1)$$

where $L_{t,c}$ is the likelihood of threat t materializing against class c and $I_{t,c}$ its class-specific impact, both over the same stated horizon. The scores of Table 1 stand in for the products $L_{t,c}I_{t,c}$ in an ordinal sense only: they expose the shift in the adversary set and its direction, and they do not constitute a calibrated expected loss, a probability, or a warrant for arithmetic across categories. The equation’s role is to make each term explicit so that the comparison can be challenged term by term.

3.1 Adversary Objectives and the System Boundary

Two design decisions are often conflated and must be separated. Choosing a lake over the sea closes the direct maritime access path; it does not, by itself, reduce overall risk or the probability of service interruption. Choosing submerged modules over land buildings complicates physical access to the modules; it does not protect the service, because the service depends on everything else. Table 2 separates the adversary’s objectives against the assets that actually carry them.

The system boundary for everything that follows includes the shore base, the modules, the power feed, the umbilical, the backbone link, the operators, the administrative access layer, and the replication links between sites. A domestic location is necessary for jurisdictional control and is not sufficient for operational control; insider management and key control must be specified inside the boundary, not assumed outside it. The design requires a visible, defended shore base, and the residual observability of that base, together with the observability of cable corridors, surface vessels, and human activity around the site, is a cost the design accepts; direct optical observability of the module itself may be reduced, while thermal, acoustic, and infrastructure signatures remain to be quantified by testing.

The strategic context sharpens the case without proving it. Southeast Asian states conduct their foreign policy under sustained great power rivalry [25], and the South China Sea episodes show how quickly maritime disputes constrict an archipelagic state’s options [3]. Energy and data infrastructure are now read as instruments of geopolitical competition [55]. A state that wants strategic autonomy in this environment should want its most sensitive computation in places where autonomy is not negotiable; inside its own land is such a place, and the question this paper tests is whether under its own water adds anything beyond it. The Indonesian doctrine of national resilience remains the operative frame, and the literature on it in-

cludes substantive criticism that must inform rather than decorate any appeal to it [33].

4 System Design and Service Definition

4.1 Architecture

The system has three physical elements. A shore base, built on the lake margin, houses power conversion, the fiber point of presence, security, and the operations staff. Sealed modules sit on the lakebed at a locally shallow bench, in the case study at 300 m depth, below the thermocline and well above the deepest basin floor; the existence of a suitable bench is a survey question for each site, not an inference from maximum depth. An umbilical bundle, armored power and fiber, joins them. The module carries compute, storage, and the internal cooling loop; the design is lake-cooled without an in-hull chiller, using powered circulation, because the lake provides the heat sink and the heat exchangers sit on the hull exterior, but pumps and fans remain active loads, and the term “passive cooling” is therefore avoided. It is not passive in the operational sense; servers, disks, and pumps are failure-prone components that live inside the hull, and the shore base concentrates the power conversion, network termination, and security functions that carry the operational exposure. Retrieval by barge is the intended mechanism for module-level maintenance, which makes the division defensible only if retrieval is demonstrated to be routine in practice, which no trial has yet shown. The electrical protection follows the principle shown in the lower panel of Figure 3: a shore-side breaker isolates the lake feeder on fault, and a module-side input isolator prevents the local UPS or any alternate path from energizing a damaged cable segment, with no automatic reclosing on a suspect feeder; residual-current detection and continuity supervision are engineered in rather than assumed. Figures 2 and 3 show the arrangement.

4.2 Materials and Ecological Discipline

The module sits in a lake that may be anoxic and is certainly stratified at the target depth. Three design constraints follow. The hull must be inert: concrete, food-grade polymers, or certified marine steel, with no copper-based antifouling and no leaching coating of any kind; the labels food-grade and marine-grade are not themselves qualifications, and the coating on Figure 2 must be specified and tested against the no-leaching rule. Fresh water is chemically milder than seawater in its chloride load, so the corrosion problem changes character; it does not automatically become easier, because local water chemistry, galvanic couples, and microbial activity all remain to be characterized. The second constraint concerns heat. The third is acoustic hardening: disk drives and other elec-

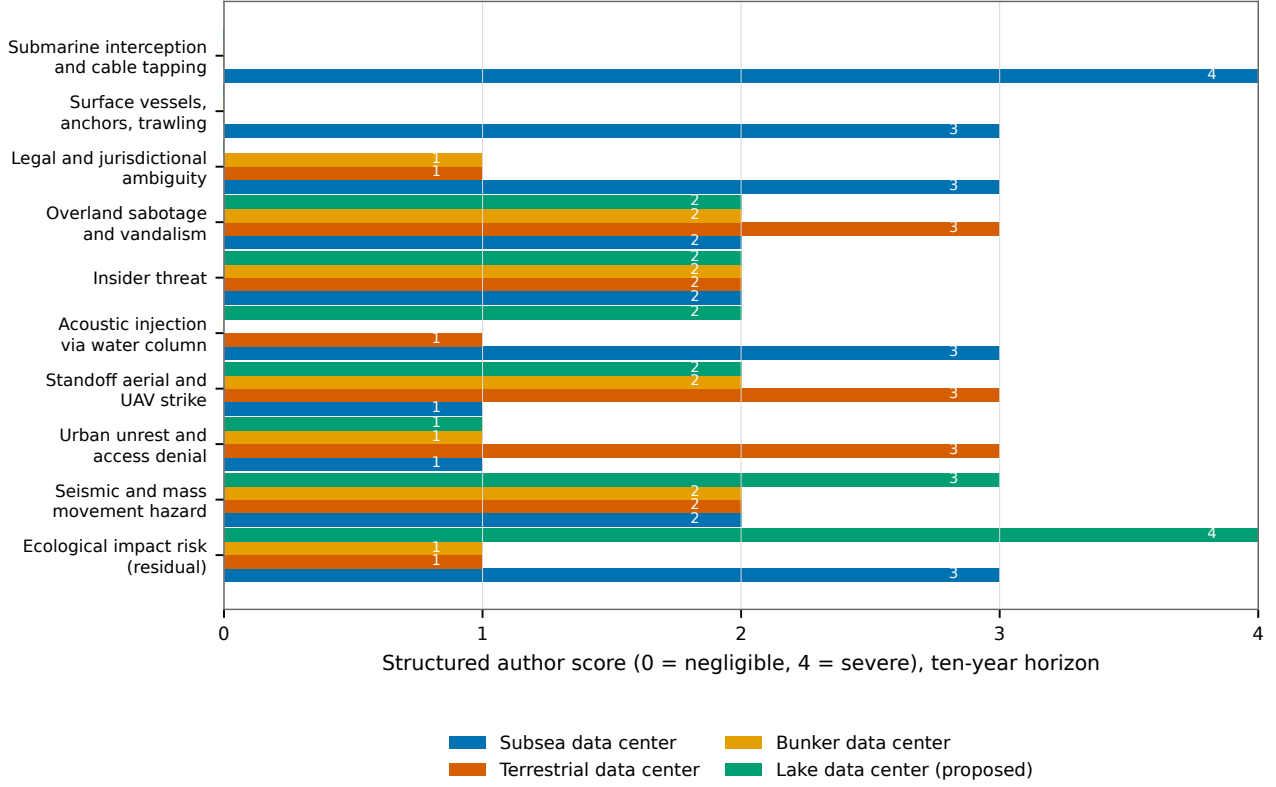


Figure 1: Structured author scores of threat exposure across four deployment classes, ten-year horizon, ordinal zero-to-four scale (zero = negligible on this scale, not absence of risk). The full matrix is published in Table 1; scores are judgments to be replaced by calibrated threat assessments, not measurements.

tromechanical components must sit on vibration-isolated mounts specified by measured transfer functions, and resonance modes of the hull–mount–component chain must be characterized, because the water column admits acoustic attack without any breach of the hull [43]; a “shielded enclosure” means mechanical damping with quantified attenuation, not an assumed analog of electromagnetic shielding. The ecological constraint is the binding one. The ecological literature shows that heat rejection into a lake alters stratification and circulation [56], and ancient tropical lakes such as Matano and Towuti hold endemic species and rare geochemical conditions that no deployment may gamble with [10, 47]. The design therefore imposes a thermal neutrality budget: the rate of heat rejection must not exceed what the hypolimnion can absorb without a change in the density structure that exceeds stated detection thresholds,

$$Q_{\text{reject}} \leq Q_{\text{budget}} \approx \rho c_p V_{\text{mix}} \frac{\Delta T_{\text{max}}}{t_{\text{fill}}}, \quad (2)$$

where V_{mix} is the hypolimnetic volume available for heat uptake, ΔT_{max} is the permitted temperature rise that preserves the density gradient, and t_{fill} is a heat accumulation timescale. Equation (2) is a dimensional framing, not a finished safety criterion. For a lake whose deep water never fully overturns, no calendar mixing period can be

assigned, and with an unbounded t_{fill} and no modeled heat export the allowable continuous input tends to zero; a defensible budget therefore requires either an explicit design horizon with a modeled export pathway, or a spatially explicit plume–intrusion model with measured thresholds, which is the program of Section 6. The budget favors low-power-density modules of the intelligence class, tens to hundreds of kilowatts, over hyperscale clusters; the lake is the limiting resource, and the design treats the lake as the binding constraint rather than the market.

4.3 Service Definition

Cost and availability comparisons are meaningless without a service to compare, and power draw is not service output: two designs at equal kilowatts can differ in throughput, storage, latency, and users. The service is defined as follows, with the caveat that the workload anchors must be fixed quantitatively at the pilot stage. *Workload:* national security and intelligence processing, moderate compute intensity, batch and interactive classified workloads; representative service targets to be fixed per application owner before pilot. *Capacity:* 100 kW of IT delivered by two 50 kW modules per site; a degraded mode of 50 kW is acceptable for up to 30 days after the loss of one module, after which the spare module must be installed; degraded-mode capacity is a target maximum, not a guaranteed

Table 1: Threat score matrix underlying Figure 1. Author judgments on a zero-to-four ordinal scale for a ten-year planning horizon, with unquantified uncertainty; the values are structured estimates for comparing deployment classes term by term, not calibrated expected losses.

Threat category	Subsea	Terrestrial	Bunker	Lake
Submarine interception and cable tapping	4	0	0	0
Surface vessels, anchors, trawling	3	0	0	0
Legal and jurisdictional ambiguity	3	1	1	0
Overland sabotage and vandalism	2	3	2	2
Insider threat	2	2	2	2
Acoustic injection via water column	3	1	0	2
Standoff aerial and UAV strike	1	3	2	2
Urban unrest and access denial	1	3	1	1
Seismic and mass movement hazard	2	2	2	3
Ecological impact risk (residual)	3	1	1	4

Table 2: Adversary objectives against system assets. Depth effects are hypotheses to be tested by scenario analysis, not established results.

Asset	Adversary objective	Threat paths	Effect of depth	Remaining dependencies	Evaluation
Modules and storage media	Seize hardware or read data at rest	In-water access, barge theft, retrieval interception	Hypothesis: raises access cost and time	Maintenance crews, insiders, shore base	Scenario assessment, controlled trials
Compute service	Deny or degrade service	Power cut, fiber cut, sabotage of shore base	No protection of the service itself	Shore base, umbilical, backbone, operators	Dependency and failure analysis
Data and system control	Unauthorized logical access	Administrative or network access	Submersion does not address this path	Identities, keys, operators, replication	Access control audit, red team
Data confidentiality	Read or leak replicated data	Cross-site replication, custody gaps, key compromise	Submersion does not address this path	Replication topology, key management, compartmentation	Need-to-know audit, key custody review

throughput. *Replication*: synchronous replication between a pair of sites with a witness at a third site for fencing and split-brain prevention, following the majority-quorum logic of consensus systems [34]. Under synchronous commit, committed-write loss is targeted at zero for the failure classes the protocol handles; a recovery point objective of 15 minutes applies to the asynchronous fallback path that activates in degraded mode, so that the phrase “synchronous with RPO 15 minutes” has an exact meaning rather than an ambiguous one. *Recovery*: a recovery time objective of 4 hours, defined as a target maximum, to degraded mode after a single-module loss, and a design target of 14 days to full capacity via module swap; these are targets to be demonstrated, and a monthly-resolution model cannot validate hour-scale RTO. *Availability*: two criteria are modeled in Section 9. The capacity criterion counts the service as available when at least two of four sites are up; the quorum criterion, which is the one a consistency-safe writable service must satisfy, requires at least three of four voting sites or an equivalent witness-fencing topology. These definitions are inputs to every calculation that follows and can be substituted by readers who wish to reproduce the results under different service assumptions.

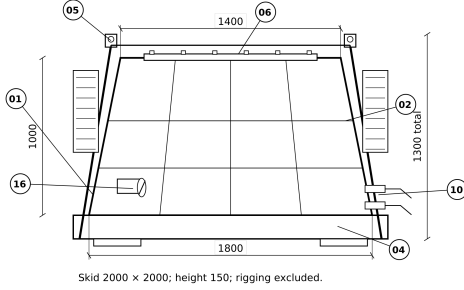
4.4 Power

Indonesian deep lakes co-locate with hydropower in a way coastal waters do not, and the national renewable energy inventory documents the resource base [19]. The submerged data center patent of [41] couples modules to wave, tidal, and temperature-differential power; the lake forfeits wave and tide but gains a firm hydropower cascade, which is the better trade for a defense customer, provided the cascade’s firmness is contracted rather than assumed. Proximity to a plant, however, does not by itself prove available capacity, connection rights, tariff, or supply priority; all four are assumptions pending negotiation with the utility, and the cost model of Section 7 treats tariff as a separated variable precisely because it is an assumption. Floating solar on the lake surface is a credible supplement, with a worldwide potential that is well documented [60], but it carries its own ecological risks and competes for surface area [9, 2]; the design places photovoltaics on the shore base roof and on disturbed land, not on the lake. Micro-hydropower on outlet rivers offers a tertiary local option [7]. Climate variability will stress hydro output over the coming decades [28], which argues for the shore-based solar supplement and for on-site storage.

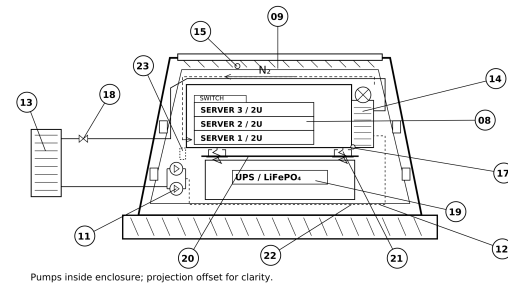
TRUNCATED SQUARE-PYRAMID UNDERLAKE DATA CENTER STRUCTURE

3 x 2U servers | Backup battery | Vibration isolation | Upper lifting points | Proposed dimensions in mm

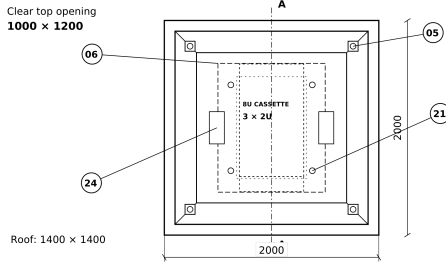
(a) Front elevation



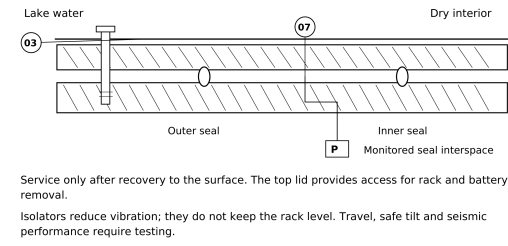
(b) Section A-A



(c) Plan / roof framing



(d) Lid seal and shell detail



- 01 Stainless-steel shell; grade TBD
- 02 Structural frame / stiffeners
- 03 Corrosion-protection coating
- 04 Ballast skid and foundation
- 05 4 upper padeyes on load-bearing frame
- 06 Watertight top service lid
- 07 Dual seals + monitored interspace
- 08 3 x 2U servers + SSD/NVMe / 8U cassette

- 09 N₂ atmosphere (OEM approval required)
- 10 A/B power / fibre-optic feedthroughs
- 11 Coolant pumps A/B
- 12 Closed coolant circuit
- 13 External heat exchangers A/B
- 14 Gas-to-liquid coil + fans
- 15 T/RH/O₂/pressure sensors + IMU
- 16 Camera + window + wiper

- 17 Leak collection tray / detectors
- 18 Coolant isolation valves
- 19 UPS + LiFePO₄ battery (optional)
- 20 Battery barrier / cooling bay
- 21 4 captive isolators + snubbers
- 22 Flexible connections + bonding
- 23 Transport / service locks
- 24 PDU / safety controls

Figure 2: Conceptual truncated square-pyramid underlake module: front elevation, section A–A, roof plan, and lid seal detail. Components include the stainless-steel shell (01), structural stiffeners (02), dual lid seals with monitored interspace (07), nitrogen atmosphere (09), A/B coolant pumps (11), A/B external heat exchangers (13), server cassette (08) on vibration isolators (21), and power distribution with safety controls (24). This is a prototype-scale engineering concept, not a qualified 50 kW unit: wall thickness, rated external differential pressure, mass, battery fault management, and the server bill of materials that fills the 50 kW envelope remain open design inputs.

4.5 Connectivity

An inland site depends on terrestrial fiber. The Palapa Ring backbone reaches the provinces where candidate lakes sit, and the shore base requires dual geographically separated fiber routes plus a microwave link and a satellite terminal for degraded-mode operation; provincial backbone reach does not by itself prove two independent routes to the tapak, which must be engineered and priced per site. The threat to inland fiber is excavation, vehicle strikes, and land disputes; route diversity and burial depth reduce these risks without eliminating them. The comparison with submarine cable vulnerability, which requires permanent sensor networks to monitor [6], plausibly favors the inland option on cost, but no bill of materials for either route class is presented here, so the claim remains a hypothesis. One honesty belongs here: the two fiber paths in the umbilical share one physical corridor between the module and the shore base, and a single event there cuts both. The design must therefore separate two different failure surfaces. First, shore-side backhaul: microwave and satellite protect the shore base's uplink if terrestrial backhaul fails. Second, the module link: no radio path

connects the shore base to a module whose umbilical is severed, and the module UPS supports only local shut-down or limited local operation, not long-term service. The response to a severed umbilical is failover of user service to the second site holding replicas and sufficient capacity. The minimal scenarios to be modeled per site are: a single fiber cut, a full corridor cut, a land-side grid outage, loss of the shore base, loss of two sites in one region, and an inter-region partition; for each, the design must state which resources remain, which communication paths survive, the data state, and the fallback procedure.

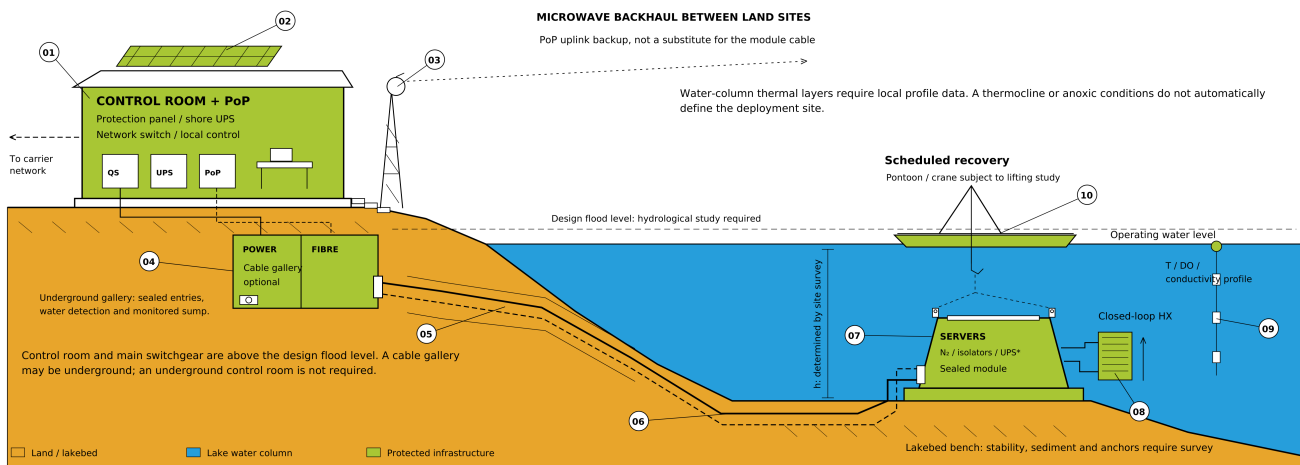
4.6 Maintainability

A lake module can in principle be winched to the surface, serviced on a barge, and redeployed; whether this takes days is a claim that only retrieval trials can establish, and Project Natick's ocean retrieval shows that lifting submerged modules is possible without showing that it is cheap or routine [31]. Retrieval converts underwater maintenance from a sealed-system problem into a civil-engineering one, and the design therefore favors modular replacement of entire sealed units over in-situ repair; the

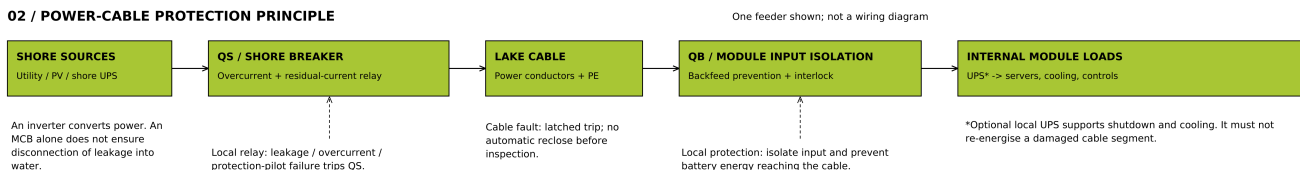
UNDERLAKE DATA CENTER / SITE SECTION AND POWER PROTECTION

Not to scale | Routes, depth, flood level and ratings remain unassigned

01 / LAND - LAKE SECTION



02 / POWER-CABLE PROTECTION PRINCIPLE



- 01 Control room / switchgear / PoP
02 Rooftop PV, supplementary source
03 Land-based radio backhaul
04 Underground cable gallery
05 Protected landfall and duct
06 Power / fibre on surveyed route
07 Truncated square-pyramid module
08 HX, no normal coolant discharge
09 Lake-condition monitoring
10 Lifting / recovery access

Solid route: power. Long-dashed route: fibre. Short-dashed rigging: recovery operation. Diagram routes do not establish geographical A/B diversity.

Concept, not a construction design. Earthing, RCD/relay type, breaker rating, trip time and habitat protection require site-specific calculation and testing.

Figure 3: Conceptual site section of an underlake data center linking shore-based control and electrical facilities, an optional cable gallery, a protected landfall corridor, and a truncated square-pyramid module on a lakebed bench. The lower panel illustrates the power protection principle: shore-side feeder disconnection and module-side backfeed prevention with inhibited automatic reclosing following a fault. The figure correctly implies the boundary of redundancy: radio links protect the shore base uplink, not the module umbilical. Cable routes, flood elevation, operating depth, stratification, and protection settings require site surveys and design analysis.

marginal cost of retrieval is a design assumption pending a time-and-motion trial.

5 Candidate Lakes and Multi-Criteria Site Selection

Indonesia has hundreds of lakes, but the screening logic eliminates most immediately. Elimination thresholds are applied before ranking, and their rationale must be visible: depth of at least 100 m, so that a module sits below the photic zone and the mixing surface layer; access to a transmission line within roughly 20 km and a backbone point of presence within roughly 50 km, because energy and connectivity are prerequisites, not weighted criteria; and exclusion of sites inside strict conservation designations, to be verified against current zoning instruments rather than assumed. Of the seven candidates in Table 3, five pass the depth filter and are ranked; Sentani (52 m) and Rawa Pening (10 m) fail it and are excluded from ranking.

Table 3: Candidate lakes and the depth filter. Depth and area are approximate publicly reported values; sources are given per row category below the table, and every depth requires bathymetric verification before use.

Lake	Depth m	Area km ²	Origin	Hydro	≥100 m
Matano	~590	~164	tectonic	Larona	pass
Toba	~505	~1130	caldera	Asahan	pass
Poso	~450	~323	tectonic	Poso	pass
Singkarak	~268	~108	tectonic	Singkarak	pass
Towuti	~203	~561	tectonic	Larona	pass
Sentani	~52	~104	tectonic	limited	fail
Rawa Pening	~10	~18.5	semi-nat.	Tuntang	fail

Sources: lake depths and origins from [37, 11, 57, 47]; regional geology from [36, 23, 46, 29, 18, 54]; energy access and zoning status are screening-level judgments pending site verification.

Depth figures are raw, publicly reported values; every entry requires bathymetric verification, and a maximum depth does not by itself establish the existence of a stable, accessible, fault-free bench at the operating depth.

The depth filter produces a geographic finding worth

stating within its scope. Within the candidate set screened here, Java, where governance capacity and backbone connectivity are strongest, offers no freshwater lake approaching the 100 m threshold; this is a statement about the screened candidates, not an inventory claim about every water body on the island. Rawa Pening, the Java candidate in the national priority lakes list, has reported depths of 0.35 to 10 m with a mean of 2.17 m and a storage volume of 48.15 million m³, against heavy sedimentation and water hyacinth encroachment [24]. Such depths provide neither concealment nor access denial, so the framework’s output for operational Java deployment is terrestrial or bunker. The shallow depth, however, confers a different value: Rawa Pening can serve as the shallow-water integration testbed of the development program, close to Semarang, to the grid, and to the Tuntang hydropower chain, allowing module, umbilical, and retrieval procedures to be exercised before any deep-lake pilot, subject to the same ecological and permitting scrutiny as any other site. The lake option is thus, on the screened candidate set, a Sulawesi–Sumatra instrument for operations with a Central Java testbed for development.

The ranking model is a weighted additive multi-criteria scheme of the family used in site selection practice [20, 45]. Six criteria are scored per lake: geohazard exposure (to be minimized), energy access, backbone connectivity, ecological sensitivity (to be minimized), depth, and social acceptance. Weights are set by the security priorities of this paper: geohazard 0.25, energy 0.20, connectivity 0.15, ecological sensitivity 0.15, depth 0.10, social acceptance 0.15. Scores are normalized to the unit interval; for depth, the only criterion with reported raw values, the normalization is the threshold-anchored linear rule

$$s_{\text{depth}} = \frac{d - d_{\min}}{d_{\max} - d_{\min}}, \quad d_{\min} = 100 \text{ m}, \quad d_{\max} = 590 \text{ m}, \quad (3)$$

which anchors zero utility at the eligibility threshold and unit utility at the deepest eligible lake, and is monotonic in reported depth. The composite score is

$$S_i = \sum_{k=1}^6 w_k s_{ik}. \quad (4)$$

The full input matrix is published in Table 4 so that the ranking can be reproduced and challenged term by term. Depth derives from the reported bathymetry of Table 3 through Eq. (3); all other criteria are ordinal author judgments on a zero-to-one scale, informed by the cited literature, and are not measurements. Figure 4 places the candidates on reported depth and geohazard exposure, and Figure 5 plots the resulting composites. Social acceptance is scored from identifiable pressures: Toba is a super-priority tourism destination, the Sorowako area around Matano and Towuti carries mining-related social friction, and Sentani sits amid customary land claims; these are qualitative judgments that structured stakeholder elicitation must replace, and they carry no quanti-

Table 4: Full scoring matrix for Eq. (4). One is favorable in every criterion; geohazard and ecology are inverted. Provenance: depth from reported bathymetry via Eq. (3); all other scores are author judgments pending measurement or elicitation. Sentani and Rawa Pening fail the depth filter and are not ranked.

Lake	Geo.	Ener.	Conn.	Ecol.	Dept.	Soc.	S_i
Matano	0.55	0.98	0.88	0.60	1.00	0.83	0.78
Poso	0.50	0.88	0.75	0.75	0.71	0.80	0.72
Towuti	0.55	0.98	0.78	0.52	0.21	0.73	0.66
Singkarak	0.42	0.85	0.85	0.60	0.34	0.55	0.61
Toba	0.55	0.88	0.65	0.42	0.83	0.25	0.59

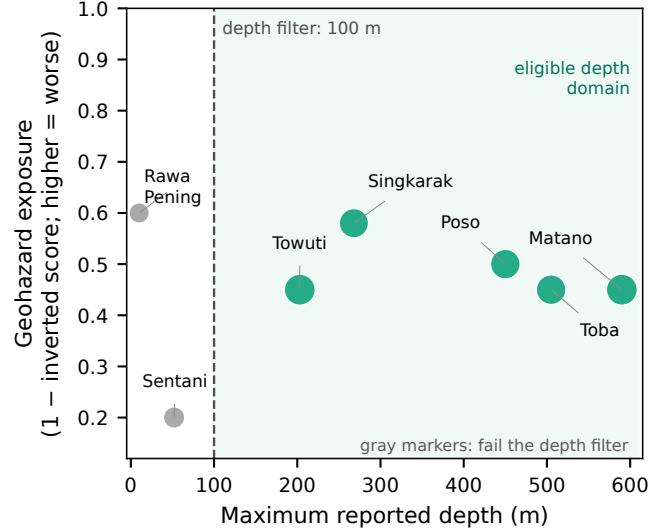


Figure 4: Candidate lakes on reported depth and geohazard exposure (the inverted scores of Table 4); bubble area indicates co-located energy access. The 100 m depth filter and the eligible domain are shown; gray markers fail the filter and are excluded from ranking.

fied uncertainty.

5.1 Sensitivity of the Ranking

Any defensible ranking requires sensitivity analysis, and stability against weight changes says nothing about stability against score error. The standard test perturbs one weight at a time while holding the scores fixed, renormalizing the weights after each perturbation. Here each weight is moved to 50% and 150% of its nominal value, twelve perturbations in all, with the scores of Table 4 held fixed. The result of this specific procedure on this specific input matrix is that the top three lakes, Matano, Poso, and Towuti, keep their ranks and membership in every one of the twelve perturbations, while the fourth and fifth ranks, Singkarak and Toba, separated by 0.015, exchange places in three of the twelve cases. The direction of each perturbation is data-dependent: lowering the geohazard weight raises Matano’s composite by 0.033, because Matano’s geohazard score of 0.55 lies below its composite of 0.78;

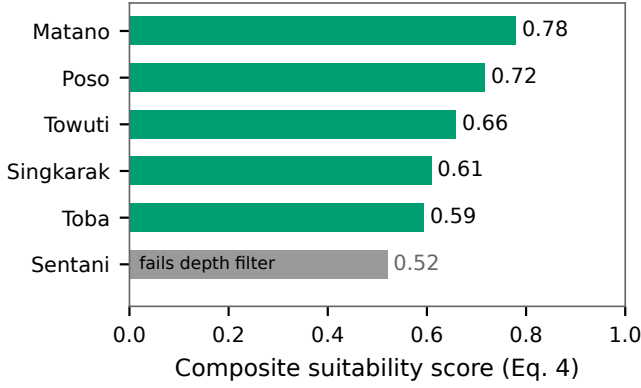


Figure 5: Composite suitability scores from Eq. (4) using the published matrix of Table 4. The gray bar shows Sentani’s score for reference; the lake fails the depth filter and is not ranked.

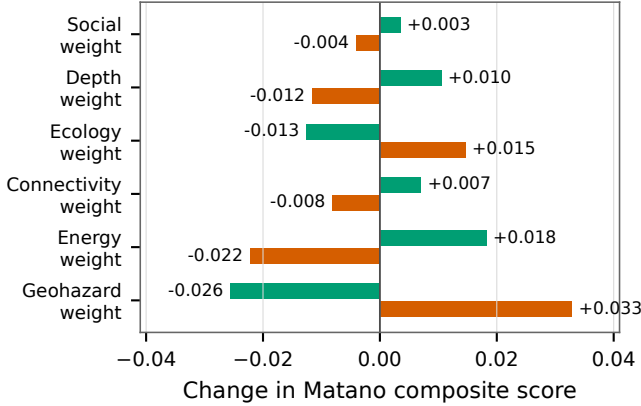


Figure 6: One-at-a-time weight sensitivity of the Matano composite score. Each bar is the score change when one weight moves to 50% (red) or 150% (green) of nominal, with renormalization. The top three ranks of Table 4 are unchanged in all twelve perturbations; ranks four and five exchange places in three.

the sign of any weight effect depends on the criterion score relative to the composite, and Figure 6 shows the full pattern. Weight robustness must not be overread. The Matano–Poso margin is 0.063, and a uniform score perturbation of ± 0.031 applied to every criterion in opposite directions would reverse the top rank; all such perturbations lie inside the zero-to-one score domain. This is an illustration of score-error sensitivity, not an empirical probability, and it means the published input matrix cannot alone distinguish the top two lakes at the precision the decimals suggest. The honest conclusion is that Matano is the priority candidate under the published screening assumptions, not an empirically confirmed best site, and that the unit of final selection is not a lake but a set of site coordinates with a cable route and a shore base, which only the Phase 0 surveys of Section 11 can produce.

The ecological stakes are concrete and must not be mini-

mized. Matano and Towuti are ancient lakes with endemic fauna [10], geochemically extraordinary sediments [57, 37], active methane cycling in ferruginous water [47], and measurable water quality dynamics [14, 48]. The geologic setting is equally concrete: the Palu-Koro fault system produced the destructive 2018 Sulawesi earthquake and tsunami [46, 54, 29], the Matano fault system runs adjacent to the first-ranked lake, and ground motion studies around the Matano fault already exist for design use [18]; the 2018 sequence is regional context, and Matano-specific hazard parameters must come from site studies rather than from Palu. Toba is a caldera system with its own resurgent history [36, 23]. These references define the two binding constraints, ecology and seismicity, that the design must satisfy simultaneously; they do not establish that either constraint is satisfiable at the ranked sites.

6 Screening-Level Thermal Calculations

No measurements from the proposed 300 m, 100 kW configuration exist, because no such facility exists; lake trials and limnological measurements exist and constrain the design, but they do not substitute for site data. What can be done now is screening-level physics, and it should be done openly, because it is the method that measurement will later replace. The case is the Matano module of Section 8: 100 kW of computing with a facility input of 120 kW at a PUE of 1.20, of which 108 kW enter the lake and the remainder is rejected to air at the shore base. The PUE of 1.20 is a scenario input, not a design result: it requires the absolute water temperature at operating depth, heat-exchanger sizing, pump and fan curves, cable losses, and the parasitic loads of the shore base, none of which is available yet. The heat enters the lake through a 100 m external heat-exchanger surface, a closed loop that exchanges no mass with the lake; “diffuser” denotes this heat-spreading surface, not a water discharge. The module sits at 300 m depth, below the thermocline, in a hypolimnion of order $4 \times 10^{10} \text{ m}^3$; both the bench and the volume are assumptions pending survey. The parameters of the calculation are published in Table 5 because a thermal argument without its parameters is unfalsifiable. Three distinct questions are asked, and each has its own physics; the basin and the intrusion lens are two views of the same rejected heat at different scales, not two independent reservoirs each receiving 100% of the load.

First, the far field. Heat rejected anywhere in the lake eventually warms the basin as a whole under the no-export assumption. The basin-average anomaly after time t is

$$\Delta T_{\text{basin}} = \frac{Q t}{\rho c_p V_{\text{hypo}}}, \quad (5)$$

which for the stated numbers gives $1.0 \times 10^{-3} \text{ K}$ after fifty years of continuous operation. The basin-average

Table 5: Parameters of the screening thermal calculations. $\rho c_p = 4.18 \times 10^6 \text{ J}/(\text{m}^3\text{K})$; thermal expansion coefficient $\alpha = 2.57 \times 10^{-4} \text{ K}^{-1}$ (assumed constant); heat-exchanger line length $L = 100 \text{ m}$; lens thickness $h = 10 \text{ m}$ (assumption); hypolimnion volume $4 \times 10^{10} \text{ m}^3$ (assumption); line-plume entrainment constant $c = 0.15$ (order of magnitude, after [51]). A 365-day year is used consistently.

Parameter	Value
Facility input (assumed)	120 kW (100 kW IT, PUE 1.20)
Heat to lake, Q	108 kW
Heat to air (shore base)	12 kW
Depth of source	300 m
Rise to thermocline base, z	200 m
Heat-exchanger line length, L	100 m
Lens thickness, h	10 m (assumed)
Hypolimnion volume	$4 \times 10^{10} \text{ m}^3$ (assumed)

effect is small under the stated fifty-year, whole-basin, no-export assumptions; it is not “negligible at any planning horizon,” because the model is linear in time and contains no mixing, no seasonal cycle, and no export, and it is not a local safety statement, because the basin mean says nothing about the near field.

Second, the near field. The module’s heat leaves the 100 m line as a buoyant plume. For a line source the entrained volume flux at height z above the source scales as $Q_v(z) = cB^{1/3}L^{2/3}z$ with buoyancy flux $B = g\alpha Q/(\rho c_p)$ [51]; the axisymmetric form $Q_v \propto B^{1/3}z^{5/3}$, which applies to a point source, is not used, because the source here is a distributed line. The temperature anomaly delivered at the base of the thermocline is then

$$\Delta T_{\text{plume}} = \frac{Q}{\rho c_p Q_v(z)}. \quad (6)$$

For a rise of 200 m, $B = 6.5 \times 10^{-5} \text{ m}^4/\text{s}^3$, $Q_v = 26 \text{ m}^3/\text{s}$, and the computed anomaly is $9.9 \times 10^{-4} \text{ K}$. Entrainment over two hundred metres of rise is an extremely effective mixer under this idealized model. Two caveats bound this result. The plume law is an unstratified theory, and its validity inside a strongly stratified metalimnion, where the plume slows and spreads, is exactly where it is least reliable; the calculation above deliberately applies it only up to the base of the thermocline, and stratification along the 200 m path, which can alter entrainment and neutral buoyancy, is not modeled. And the result says nothing about the concentrated near-field region within tens of metres of the heat-exchanger line, which requires a separate diffuser design calculation.

Third, the intrusion. Warm water trapped below the thermocline spreads laterally beneath the density step. Rather than asserting a physical lens whose dynamics the model does not resolve, we report a heat-equivalent radius: the radius of a hypothetical cylindrical volume of prescribed thickness h and prescribed temperature anomaly ΔT_{lens} that would contain the heat rejected in time t ,

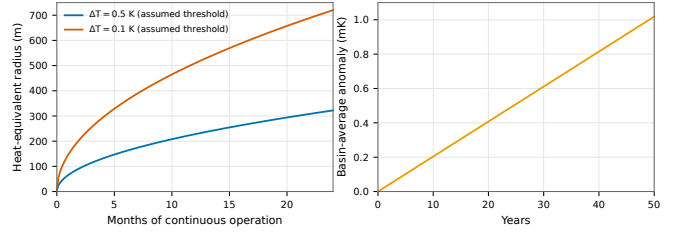


Figure 7: Left: heat-equivalent radius beneath the thermocline for two prescribed detection thresholds (Eq. (7)); the model contains no growth-arrest mechanism, so the curves are screening indicators, not physical bounds. Right: basin-average temperature rise over fifty years (Eq. (5)).

$$r(t) = \sqrt{\frac{Q t}{\rho c_p \pi h \Delta T_{\text{lens}}}}. \quad (7)$$

The anomaly is a prescribed detection threshold of an instrumented monitoring program, 0.1 K and 0.5 K, not a prediction of the physical lens anomaly; the plume dilution of Eq. (6) does not feed Eq. (7), and to make the point explicit, using the plume anomaly of $9.9 \times 10^{-4} \text{ K}$ in Eq. (7) would produce a radius of about 5 km, which demonstrates that the two quantities are different things and must not be chained. With $h = 10 \text{ m}$ as a stated input, the 0.5 K heat-equivalent radius reaches roughly 228 m after twelve months, and the 0.1 K radius roughly 509 m. Equation (7) is a heat balance and contains no mechanism that stops growth; the radius grows as the square root of time without limit in the model, so it is a monitoring-scale indicator, not a physical bound, and the actual isotherm geometry, thickness, and intrusion depth require a calibrated hydrodynamic model. Across the uncertainty envelope, Q from 54 to 216 kW and h from 5 to 20 m, the twelve-month 0.5 K radius spans 114 to 456 m (365-day year); Figure 7 shows both indicators. The instrumented pilot of Section 11 must measure this footprint with thermistor strings and tracers, and its readings, not this indicator, will define the ecological footprint.

The falsification conditions are stated so that the pilot constitutes a genuine test, and they must be operational, which means thresholds with durations and pre-registered decisions, not sensor resolution alone. A sensor that cannot resolve a change is not evidence of safety, and a worse sensor does not enlarge a safe budget. The predictions to be checked are: thermistor strings near the heat exchanger and far from it, with a documented seasonal baseline and control sites, show no temperature change exceeding the pre-registered threshold and duration; the density-gradient structure, dissolved oxygen, and conductivity at the thermocline remain within the baseline envelope; the plume dilution is consistent with Eq. (6) to within the stated assumptions; and the heat-equivalent footprint grows on the modeled order and not beyond it. Any measured violation of the thermal budget triggers immediate power shutdown, which is faster than retrieval, followed by the retrieval plan;

the ecological endpoint of concern is the density structure and its biogeochemical consequences, of which temperature is a proxy, not the whole story [17, 56]. The honest limitation belongs in the same paragraph as the results: these are order-of-magnitude screening calculations with idealized geometry, no bathymetry, no currents, and no seasonal cycle. Design-grade work requires a calibrated three-dimensional hydrodynamic model of the actual lake, the class of model that produced the thermal pollution findings of [56], driven by a field campaign. The method of this section is the first rung of that ladder.

7 Comparative Assessment: Terrestrial, Lake-Cooled Terrestrial, Bunker, and Lake

The lake option should be judged against its real alternatives at identical service, not against the idea of a data center in the abstract. The isolated research question is: after domestic jurisdiction, remote siting, energy proximity, and multi-site redundancy are held equal, what does submersion add, and at what cost? The comparison fixes the service of Section 4, 100 kW of defense-grade computing hardened against attack and seismic loading, operated for ten years, undiscounted, and models four facility classes: (A) a hardened terrestrial facility with perimeter security, seismic design, and grid power; (B) the decisive comparator, a hardened terrestrial facility at the same site with a closed-circuit lake-water cooling loop, which captures the thermal benefit of the lake without submerging the servers; (C) a mountain bunker of the Swiss pattern; and (D) the submerged lake facility designed here. Costs are bottom-up engineering estimates assembled from component line items with quantity and unit-cost judgments, carrying an uncertainty of roughly $\pm 30\%$ that is an engineering judgment, not a statistical confidence interval; no vendor quotations support these numbers. The estimates cover modules, shore base, connectivity, power, monitoring, permits, spares, batteries, and decommissioning at screening level; hardware refresh is excluded from all four options equally, with the caveat that refresh access costs differ across options and that exclusion is therefore not strictly neutral. Table 6 states the comparison.

The result requires a plain reading; Figure 8 stacks the ten-year totals. At equal grid tariff the lake facility is not the cheapest; the hardened terrestrial facility is, at 6.00, and the lake-cooled terrestrial comparator costs 6.34, so the thermal benefit of the lake without submersion costs about 0.34 million dollars and the submerged design costs 7.85, a premium of 1.85 million dollars (31%) over conventional hardening and 1.51 million over the lake-cooled ground facility. The bunker costs 8.11; the lake is cheaper than the bunker by 0.26 million dollars, a margin well inside the stated $\pm 30\%$ uncertainty, so “cheaper than

Table 6: Ten-year cost model at identical service (100 kW IT, hardened, 10 years, undiscounted USD). PUE assumptions: 1.60 terrestrial, 1.30 lake-cooled terrestrial, 1.50 bunker, 1.20 lake; the lake-cooled value reflects elimination of the chiller with the pumps, heat exchangers, and lake loop remaining powered. Energy is shown at two tariffs to separate tariff from PUE effects; the lake’s hydro-adjacent tariff is an assumption, not a verified right. All values are screening estimates with roughly $\pm 30\%$ engineering uncertainty.

Item (\$M)	Terrestrial	Lake-cooled	Bunker	FLDC
Capex	1.50	2.00	3.50	4.10
Energy, \$0.10/kWh	1.40	1.14	1.31	1.05
Energy, \$0.06/kWh	0.84	0.68	0.79	0.63
Maintenance, 10 yr	0.60	0.70	0.80	0.90
Staffing, 10 yr	2.50	2.50	2.50	1.80
Total, \$0.10/kWh	6.00	6.34	8.11	7.85
Total, \$0.06/kWh	5.44	5.88	7.59	7.43

the bunker” is a central-estimate statement, not a robust cost ranking. The general form of the cost equation is

$$\text{TCO} = C_{\text{cap}} + \sum_{t=1}^T (E_t + M_t + S_t), \quad (8)$$

and the differences lie partly, but not only, where the security properties lie; maintenance access, cooling equipment, and staffing models also differ across options. The lake pays a capex premium for the submerged module, the umbilical, and the installation, and it recovers part of that premium through energy and staffing; the properties no dollar figure contains, concealment of the module from direct imagery, physical access denial, and jurisdictional certainty, must justify the remainder, and the staffing savings of 1.8 against 2.5 million dollars need a roster showing which shore functions are covered before they are credited. Whether the remaining premium is worth paying is a defense policy judgment conditioned on engineering evidence that does not yet exist. A further comparison belongs in the same breath: a network of four sites at the same service level costs on the order of 24.0 million dollars for hardened terrestrial, 25.4 for lake-cooled terrestrial, 31.4 for submerged, and 32.5 for bunker facilities, all at the grid tariff with the same uncertainty. The terrestrial network is cheaper. The submerged network purchases diversity of failure mode, concealment, and access denial with that premium, and the premium is the quantity the policy debate must weigh. Two caveats keep this honest. First, the availability benefit of a four-site network is a property of siting and replication, available to any facility class; Section 9 models it to define the service target every option must meet, not to attribute benefit to submersion. Second, a mixed network, two terrestrial and two submerged sites, may diversify failure modes more cheaply than either homogeneous network, and modeling it is deferred to the Phase 0 design study of Section 11.

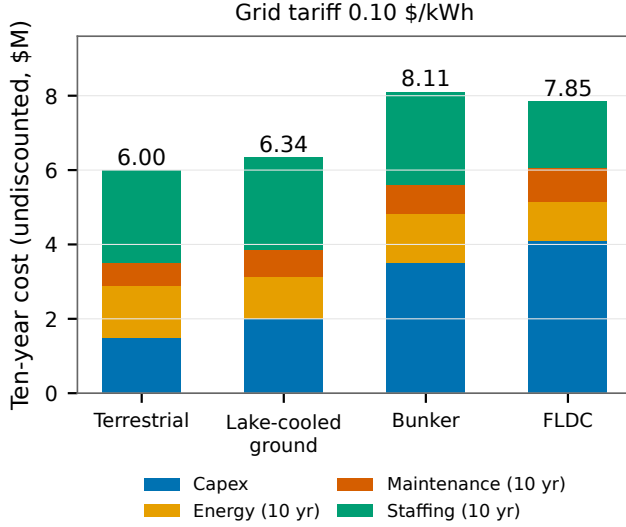


Figure 8: Stacked ten-year costs of Table 6 at the grid tariff. The submerged design is not the cheapest: the hardened terrestrial facility is, and the lake-cooled terrestrial comparator captures most of the thermal benefit without submerging the servers.

8 Case Study: Lake Matano

The case study fixes the parameters of one concrete deployment, because a design argument lives or dies on its numbers; it does not fix every parameter, and the unfixed ones are listed rather than hidden. Lake Matano is the priority candidate under the published screening: a tectonic lake of roughly 590 m maximum depth in South Sulawesi, persistently stratified in its deep basin [17, 47], with the Larona hydropower cascade nearby and the Palapa Ring within terrestrial reach. Table 7 states the design. The configuration decision is explicit: the module rests on a lakebed bench at 300 m local depth, below the thermocline, on footings, and does not float in the water column; the deepest basin floor is not used. Other candidate lakes cannot use the same operating depth, and each site requires its own bathymetry and bench selection, e.g., a Towuti deployment would seek a bench well below its thermocline within its 203 m envelope.

The pressure environment is the simplest to state and the least sufficient to qualify the structure. Hydrostatic pressure grows linearly, $p = p_0 + \rho g z$, and Figure 9 plots it; with fresh-water density 1000 kg/m^3 , gravity 9.81 m/s^2 , and surface pressure 1.013 bar, the module hull sees 29.4 bar gauge and 30.4 bar absolute at 300 m. The 60 bar rating is a design requirement for the external differential pressure, not a demonstrated qualification; the ratio of two covers neither seismic loads, lifting loads, nor fatigue. As an order-of-magnitude illustration of what the rating must carry, a flat $1.4 \times 1.4 \text{ m}$ panel of the frustum at full depth carries a resultant of about 5.77 MN, so shell thickness, stiffener layout, buckling with imperfection sensitivity, seal extrusion, and penetrator design are

Table 7: Lake Matano deployment parameters (design values; survey and test verification pending).

Parameter	Value
IT capacity	2 modules $\times$ 50 kW = 100 kW; degraded mode 50 kW
Facility input	120 kW at PUE 1.20 (assumed); 108 kW to the lake, 12 kW to air
Operating depth	300 m on a lakebed bench; basin maximum $\sim$ 590 m
Hydrostatic pressure	29.4 bar gauge, 30.4 bar absolute, at 300 m in fresh water; hull rated to 60 bar external differential (design requirement)
Heat rejection	100 m external heat-exchanger line; plume ΔT at thermocline base $9.9 \times 10^{-4} \text{ K}$
Footprint	0.5 K heat-equivalent radius $\sim$ 228 m at 12 months, $h = 10 \text{ m}$ assumed
Umbilical	4 km armored, $2 \times$ 48-core fiber, 1 MW power; single corridor
Backhaul	dual terrestrial routes + microwave + satellite degraded mode (shore uplink only)
Maintenance	quarterly ROV inspection; annual retrieval; single-module MTTR target 14 days
Capex	\$4.1 M (modules 0.7; heat rejection 0.15; shore base 1.2; umbilical 0.45; installation 0.3; monitoring 0.25; engineering 0.4; contingency 0.65)

structural problems to be solved in Phase 1, not settled by a drawing. External pressure design formulas are standard [35], but their application to this geometry, with its lids, penetrations, and pressure cycling during retrieval, requires analysis and proof testing.

Depth buys concealment and access denial, but it also sets the maintenance concept, and the failure-mode mapping must be exact to avoid double counting. Retrieval takes a barge and a winch, and the 14-day mean time to repair refers to a single-module swap with a spare on the shore base, a target to be demonstrated by a retrieval trial. The monthly-resolution availability model of Section 9 cannot represent 14 days, so independent failures there carry repair durations of one to two months, which include swap logistics and shore equipment repair; the model is conservative relative to the swap target. The mapping is: module-level failure, 14 days by swap at the design level, one month at the model’s monthly resolution; site-level damage, months by reconstruction, as in the model.

The seismic constraint is the case study’s most binding condition, and the claim is stated as a requirement, not as an achievement. Matano sits beside the Matano fault system, and ground motion studies around the Matano fault exist for design use [18]; the Palu-Koro system produced the 2018 Sulawesi sequence [46, 54, 29], which provides regional context and a warning, not a Matano hazard value. The design response is the inequality of Section 10 together with setbacks from mapped fault traces and base

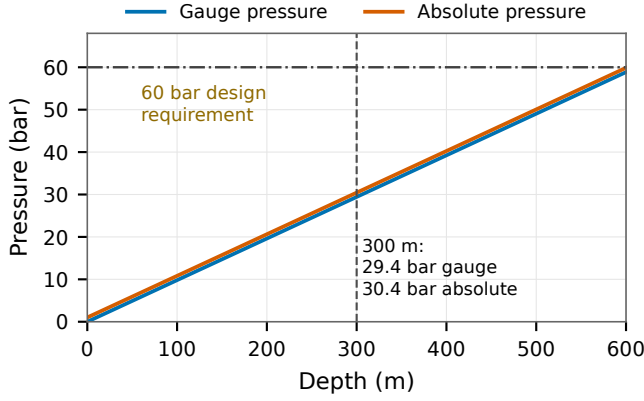


Figure 9: Hydrostatic gauge and absolute pressure versus depth in fresh water. The Matano operating depth of 300 m corresponds to 29.4 bar gauge (30.4 bar absolute); the 60 bar rating is a design requirement, not a demonstrated qualification.

isolation of the shore base. A module resting on footings on the lakebed is not decoupled from ground motion, and its seismic response, including shaking, sliding, overturning, settlement, fault displacement, slope failure, sediment flows, hydrodynamic loading, seiche, and umbilical strain, remains to be quantified; the pressure vessel overdesign of Table 7 does not substitute for that analysis. The shore base is the element on which the seismic budget is spent first.

9 Common-Mode Failure and Simulated Service Recovery

A single lake site, however good, remains a single site. The national argument for the program is the network: sites in different lakes, on different fault systems, so that no single natural or hostile event, within the modeled classes, takes the service down; no model in this paper can exclude events, such as shared software, shared administration, or backbone-level failures, that act on all sites at once. The modeled network has four sites: Matano and Poso in Sulawesi, exposed to a shared regional seismic source; Singkarak in West Sumatra and Toba in North Sumatra, sharing grid and backbone corridors partially, modeled as separate regional processes with correlated rates. The service criterion is that of Section 4, modeled in both forms: the capacity criterion, service up when at least two of four sites are up, and the quorum criterion, service up when at least three of four voting sites are up, because a two-of-four count is not by itself a consistency guarantee, two live sites can form two groups each believing it is authorized, and a majority quorum or an equivalent witness-fencing topology is what a writable service actually requires [34].

For independent, identical sites each criterion has an exact form,

$$A_{k\text{-of-}n} = \sum_{j=k}^n \binom{n}{j} A^j (1-A)^{n-j}, \quad (9)$$

which for $k = 2$, $n = 4$, and $A = 0.95$ gives 0.999519, and for $k = 3$ gives 0.985981. These are benchmarks under marginal site availability A , which is not the same quantity as a monthly failure probability; the simulation distinguishes the two. The independence assumption is false for a shared fault system, and the correlated case is the reason for simulation. The model is a monthly Bernoulli process, fully documented in the companion repository <https://github.com/jalaranu/companion> (`sim_availability.py`): 600 months and 2000 replications per scenario with a fixed seed (20260921); all sites start up with no burn-in; each month, remaining downtimes are decremented, each up site fails independently with probability p_{ind} for a repair drawn uniformly from one to two months, a Sulawesi regional event arrives with probability p_{CC} per month and disables both Sulawesi sites for two to six months, a Sumatra event arrives with probability 0.003 per month and disables both Sumatra sites for one to four months, and any event arriving while a site is already down extends the outage to the maximum of the two durations, so outages are never double counted. The service state is assessed after the monthly update. For independent Bernoulli failures the superposition of an independent and a common-cause process has the union form $p_i = p_{\text{ind}} + p_{\text{CC}} - p_{\text{ind}}p_{\text{CC}}$, which is the monthly-probability counterpart of the rate superposition

$$\lambda_i = \lambda_{\text{ind}} + \lambda_{\text{CC}, r(i)}, \quad (10)$$

valid only as an intensity statement; the marginals are the least interesting part of the model, because simultaneity is the whole point, and it is imposed by construction, one regional event removes every site in the region at once.

The baseline result with $p_{\text{ind}} = 0.05$ and $p_{\text{CC}} = 0.004$ (a Sulawesi event about once every two decades) is a capacity availability of 0.9952 (mean $\pm$ two standard errors 0.0003), equivalent to roughly 42 hours per year as a conversion of proportions rather than a prediction of individual incidents, and a quorum availability of 0.9484 (± 0.0013), equivalent to roughly 452 hours per year. Because the failure probabilities are hypothetical scenarios rather than observations, the model was swept over $p_{\text{ind}} \in \{0.02, 0.05, 0.10\}$ and $p_{\text{CC}} \in \{0.002, 0.004, 0.008\}$, nine scenarios, with results in Table 8; the capacity criterion spans 0.9797 to 0.9990 and the quorum criterion 0.8649 to 0.9797. The results are consequences of assumptions and must be quoted as such.

Recovery is the second number. Figure 10 aligns simulated Sulawesi events at month zero and plots the fraction of event windows in which each criterion holds, from six months before to twenty-four months after. The window rule is: for each replication, the first Sulawesi event with at least six months of preceding baseline and at least twenty-four months of following history inside the 600-month

Table 8: Nine-scenario availability sweep (mean over 2000 replications; standard errors are at most 3.3×10^{-4}). Sumatra common-cause probability fixed at 0.003 per month. Both service criteria of Section 4 are reported. Companion code: <https://github.com/jalaranu/companion> (`sim_availability.py`).

p_{ind}	p_{CC}	$p_{\text{CC, Sum.}}$	≥ 2 of 4	≥ 3 of 4
0.02	0.002	0.003	0.9990	0.9797
0.02	0.004	0.003	0.9985	0.9720
0.02	0.008	0.003	0.9975	0.9563
0.05	0.002	0.003	0.9964	0.9559
0.05	0.004	0.003	0.9952	0.9484
0.05	0.008	0.003	0.9930	0.9331
0.10	0.002	0.003	0.9858	0.8859
0.10	0.004	0.003	0.9838	0.8790
0.10	0.008	0.003	0.9797	0.8649

horizon is selected; replications without such an event are censored. This yields 1628 eligible windows, against a raw expectation of 4800 event occurrences across all replications, because the analysis needs one clean window per path rather than every event; the rule, like every other rule, is in the companion code. The capacity criterion falls from its long-run level to 0.84 in the event month and returns to the long-run level within about four to six months, inside the modeled repair horizon of the damaged pair; the quorum criterion falls to zero during the event, because a regional event removes two of four voters and a majority cannot form, and it is this architectural fact, not any facility property, that drives the availability difference between the two criteria. The mean $\pm$ two standard errors shown in the figure are computed across replications, respecting that windows are selected one per replication. The lesson is architectural. Redundancy across fault systems, not redundancy within one lake, is what produces the curve of Figure 10; the same curve is available to a distributed terrestrial network, and it costs the program a backbone and replication design, which is a real engineering expense, not merely siting well.

10 National Resilience and Intelligence Requirements

A data center that serves defense and intelligence functions must satisfy requirements that differ from those of commercial hosting; Table 9 maps the requirements selected for this proposed service onto design responses and verification methods. The list deserves emphasis because it reverses the usual logic: here the security requirement drives the engineering, and cooling follows.

Three points follow from the table. The first is that submersion is a physical layer defense and nothing more. The cyber layer requires the ordinary apparatus of intelligence-driven defense [49]. The second is that the lake offers genuine and rare properties to the physical layer: concealment of the module from direct imagery, complication of physical access, and a jurisdictional certainty that no

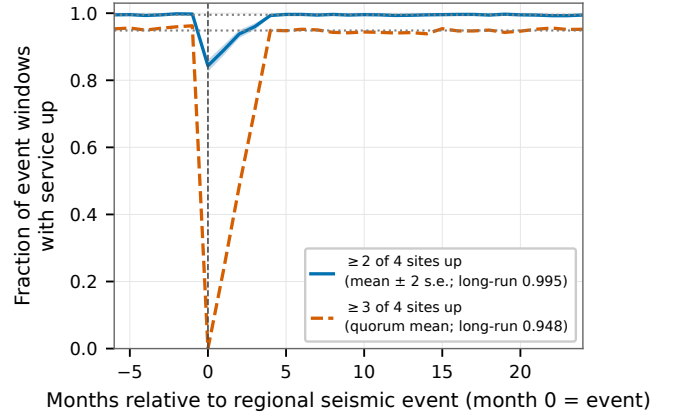


Figure 10: Simulated service recovery around a regional seismic event disabling both Sulawesi sites. Mean and mean $\pm$ two standard errors over 1628 eligible event windows (2000 replications; first eligible event per replication, six months of pre-event baseline, twenty-four months of follow-up, the remainder censored). The capacity criterion recovers to its long-run level within four to six months; the three-of-four quorum criterion is zero during the event by construction.

undersea or foreign-hosted alternative can match; how much each property is worth, and how much of it survives a competent adversary, remains to be priced and tested rather than assumed. Third, submersion complicates classical physical intrusion, but the acoustic experiments of [43] show that the water itself is an attack channel that terrestrial walls attenuate and lakes merely make observable. The defense is monitoring and isolation, not water.

The statutory regime sharpens three requirements of Table 9. First, the Masa Retensi of intelligence secrets, a protection and storage period of 25 years extendable with the approval of the House of Representatives (UU 17/2011 Pasal 1 angka 7; Pasal 25 ayat (3)–(4)), sets an information-retention duty, not a requirement that any single physical medium survive 25 years; the design response is therefore the retention architecture of Table 9, executed across the multi-site network with scheduled migration, which is equally available to terrestrial facilities and is claimed here as a design requirement, not as a proven property of sealed modules [52]. Second, compartmentation is a statutory principle of intelligence activity (Pasal 2 huruf c), not merely a security preference, and the multi-site network must implement it: replication between sites respects need-to-know boundaries rather than replicating wholesale, which the synchronous-pair topology of Section 4 must enforce at the key-management and access-control layers. Third, the secrecy duty of intelligence personnel extends by statute to the special facilities and equipment of intelligence activity (Pasal 18 huruf b), which places the low-signature module design inside, rather than outside, the legal framework, and the criminal penalties attached to disclosure (Pasal 44–45) support, in the author’s ordinal judgment, the highest impact weight for data leakage in Eq. (1); that weight is a judgment, not

Table 9: Military and intelligence requirements mapped to design features and verification.

Requirement	Design response	Verification
Territorial data residency under full legal control	Deployment in internal waters, national operator, domestic staff	Compliance audit under PP 71/2019 Pasal 20–21 and UU 27/2022 Pasal 56
Low observability of the module	Modules below the photic zone and thermal budget of Eq. (2); the shore base, cable corridors, and vessel traffic remain observable	Multispectral remote sensing trials; residual signature quantification
Physical access denial	Depth plus a single defended shore access point	Adversarial red team exercise
Acoustic attack through the water column	Hydrophone arrays, vibration-isolated mounts with measured transfer functions, mechanically damped enclosures, anomaly detection	Controlled acoustic injection trials against the actual storage bill of materials [43]
Continuity under domestic unrest	Remote siting, independent hydropower feed, dual fiber, satellite and microwave uplink for the shore base	Tabletop and field drills
Seismic survival	Fault setback criteria, base isolation of the shore base, hull design to the national seismic provisions (SNI 1726:2019); lakebed module response to be quantified	Shake table and pressure-cycling tests to site-specific ground motion [44, 18]
Forensic integrity and statutory retention	Retention architecture: integrity scrubbing, offline archival copies, scheduled migration, custody documentation, key management across time; annual retrieval and media refresh; media lifetime is not assumed	Digital forensics and custody audits; materials tests only for physical evidence preservation
Cyber protection	Orthogonal to submersion; no water barrier exists in the cyber domain	National cyber defense doctrine and threat intelligence practice [49]

a statutory number [52]. These provisions anchor in law what the cost model of Section 7 could not price.

10.1 The Interdisciplinary Frame: Ipoleksosbudhankamnas

The preceding requirements are security requirements in the narrow sense. Indonesian national resilience planning assesses strategic programs across the interdisciplinary dimensions summarized by the acronym Ipoleksosbudhankamnas: ideology, politics, economy, social conditions, culture, defence, security, and the national interest that binds them. The acronym and its dimensions are drawn from Indonesian strategic planning usage; their operational definitions in specific ministries are outside this paper’s scope. A program of this kind is not complete until each dimension has been examined, and Table 10 records the examination.

The seismic requirement is expressed as a design criterion rather than a preference, and as a requirement to be satisfied through a standard, not as a completed design. Let a_{ref} be the site-specific reference peak ground acceleration from the national hazard map, documented for the Matano fault region in the ground motion literature [18]. The design acceleration for a defense-critical module must satisfy

$$a_{\text{design}} \geq a_{\text{ref}} \gamma_I, \quad (11)$$

where $\gamma_I > 1$ is an importance factor assigned to facilities whose loss is strategically intolerable. The inequality is illustrative until the governing standard is fixed: SNI 1726:2019 provides the national provisions for seismic design, including response spectra, site classes, and impor-

tance categories, and the Phase 1 program must specify the applicable clauses, the hazard level, site effects, the vertical component, and the fragility functions of the hull and shore base rather than cite the inequality alone [44].

11 Innovation System and Technology Roadmap

Engineering feasibility being conditional does not make the deployment inevitable. A technology of this kind emerges only inside an innovation system that performs the seven functions identified in the technology innovation system literature: knowledge development, knowledge diffusion, entrepreneurial activity, guidance of the search, market formation, resource mobilization, and the creation of legitimacy [12]. Table 11 assesses each function for Indonesia. The status labels have operational definitions: emerging means documented research activity with no industrial capability; weak means some actors but no organized function; absent means no identifiable national activity in the public record, where the record is defined by the documents searched, national journals, institutional websites, and procurement listings up to September 2026; absence from that record is evidence of absence only from that record, and classified defense activity would by definition not appear. The table is the author’s preliminary diagnosis, with no interviews conducted; it is a hypothesis to be verified by structured stakeholder inquiry, not a finding.

The remedy is a gated roadmap, shown in Figure 11 and compressed to a ten-year horizon from 2026, because a capability program that cannot demonstrate value within a decade will not survive the institutions it depends on. Each phase ends in a gate with measurable acceptance

Table 10: The underlake data center program across the Ipoleksosbudhankamnas dimensions of national resilience.

Dimension	Relevance to the program	Analysis in this paper
Ideology	Technological sovereignty as a material basis of the national outlook; domestic control of strategic data	Legal framework (UU 27/2022, PP 71/2019, UU 17/2011); residency and jurisdiction analysis
Politics	Infrastructure that survives transitions of government and periods of unrest	Threat model against domestic unrest; remoteness and multi-site redundancy
Economy	Cost at equal service; energy economics; regional industrial development near hydropower	Ten-year cost model with separated tariff and PUE; co-location with the Larona and Poso cascades
Social	Local communities, demonstrations, and benefit sharing around sites	Social acceptance criterion in the ranking, as a judgment pending structured elicitation
Culture	Customary land claims and culturally significant lakes	Sentani customary context and Toba tourism designations scored in the matrix
Defence	Physical protection of strategic computing against state adversaries	Threat model; concealment, access denial, acoustic hardening
Security	Continuity of the service itself under attack, disaster, and insurgency	Common-mode failure model; simulated recovery; seismic criterion

Table 11: Innovation system functions for a lake data center program in Indonesia. Preliminary author diagnosis from public documents (national journals, institutional websites, procurement listings; September 2026); no interviews conducted; absence from the public record is not proof of absence.

Function	Status	Evidence
Knowledge development	Emerging	Limnology and seismology literature exists [37, 46]; submerged-DC engineering concentrated abroad
Knowledge diffusion	Weak	No national program; no standards body work identified
Entrepreneurial activity	Weak	No domestic submersible-DC firm identified in the search scope
Guidance of the search	Absent	No policy document in the searched set names lake deployment as an option
Market formation	Absent	No procurement pathway identified; demand concentrated in state actors
Resource mobilization	Emerging	Hydropower and backbone exist; no dedicated fund identified
Creation of legitimacy	Absent	No public engagement of security institutions identified

criteria, and preparation activities of later phases may overlap earlier ones, while no phase may proceed past its gate before the gate is passed. Phase 0, a feasibility study, completes the data gaps this paper leaves open: bathymetry at bench resolution, thermal baselines, ground motion, and ecological baselines; its exit gate is a verified site dossier with acceptance thresholds and a go/no-go decision by a named authority. Phase 1 builds a land-based prototype; its exit gate is a hull proof test at 1.5× the rated external differential, a leak test, and shake table testing of the specified test article, component or full-scale

as stated in the test plan, to the site spectrum of SNI 1726:2019. Phase 2 exercises the module, umbilical, and retrieval procedures at Rawa Pening, the shallow-water testbed of Section 5; its exit gate is a completed set of wet-deployment procedures and electrical protection drills, with no injury to personnel and no measured exceedance of habitat thresholds. Phase 3 submerges a pilot module in a deep lake under full ecological monitoring; its exit gate is the thermal falsification battery of Section 6 passed against thermistor and tracer data, with removal pre-authorized on failure. Phase 4 delivers the first operational unit serving non-classified government data; its exit gate is a full red team exercise and an operations audit. Phase 5 expands to the network of lake sites; its gate is meeting, in operation over a pre-agreed observation period with a pre-agreed confidence level, the availability criterion of Section 9, including recovery drills, because Section 9 defines the service-up rules and scenario results but does not by itself fix an operational pass target. The phases are illustrative and dated from 2026; no phase is currently underway, and the responsible institutions are not yet named because the legitimizing step has not been taken. The roadmap method itself is standard foresight practice [62, 1], and its object here is the deliberate construction of a capability, which is how transformative innovation policy frames the task [40].

12 Discussion: Evidence Status and Honest Limits

The numbers in this paper have four different statuses, and they must be kept distinct. Table 12 fixes the status of each class of input; note that author-judgment scores are ordinal inputs, not computed evidence, and that reported lake depths are not surveyed bench depths.

Four limits must be stated without softening. First, no lake data center has been built, so this paper is a design argument with screening-level calculations, not a demonstration; the results are consequences of stated assumptions, and the companion code <https://github.com/>

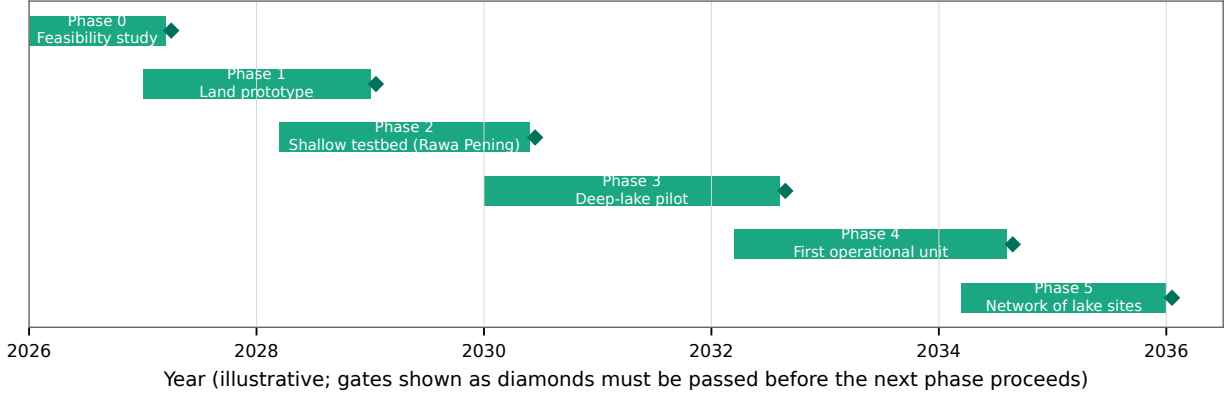


Figure 11: Gated technology roadmap from feasibility study to a national network of lake sites, dated from 2026. Diamonds mark exit gates; later phases may begin preparation before earlier gates pass, but no phase proceeds past its gate until the gate criteria are met.

Table 12: Evidence status of the inputs used in this paper.

Status	Examples in this paper
Known from sources	Reported lake depths and origins; fault context; acoustic attack results at testbed scale; statutory provisions as cited
Assumed	Tariffs, PUE values, failure probabilities, repair durations, lens thickness, hypolimnion volume, cost line items, threat scores
Computed	Ranking scores, sensitivity, thermal anomalies, costs, availability, recovery
Unproven	Field performance, ecological safety, seismic and pressure qualification, retrieval ease, residual observability

[jalaranu/companion](#) makes them recomputable, which is reproducibility of the arithmetic, not validation of the physics. Second, the ecological risk is real and partially unknown. The thermal calculations of Section 6 provide heat-equivalent indicators, not a measured footprint, and the radius equation contains no growth-arrest mechanism; the effect of a permanent heat source on the stratification of a persistently stratified ancient lake can be settled only by the instrumented pilot, which must be designed to be removed if the measurements fail, with removal itself assessed for reversibility. Third, seismic hazard at the preferred sites is severe; the engineering answer, setback, isolation, and overdesign, is a set of design requirements with a verification plan, not a demonstrated capability. Fourth, the cost analysis shows the lake option is not the cheapest way to build computing; the hardened terrestrial facility is, at equal tariff, and the lake-cooled terrestrial comparator closes most of the thermal gap at a fraction of the premium. The case for the lake rests on the security properties, and those properties have not been priced because they cannot be priced honestly until a threat assessment with access to classified data is performed; a conditional and break-even analysis of the premium

Table 13: What this paper supports and what it does not.

Supported as a framework paper	Not claimed: readiness	deployment
Reproducible calculations with published inputs and companion code	Site survey and bathymetry completed	
Novelty claims bounded by a stated search scope	Geotechnical and structural verification	
Threat model and service definition with quorum semantics	Pressure, leak, and retrieval testing	
Equal-service comparators, including the lake-cooled terrestrial option	Power and connectivity rights verified	
Open assumptions and uncertainty, with weight and scenario sweeps	Instrumented pilot and environmental measurement	
Falsification conditions, with thresholds to be fixed at Phase 0	Service targets demonstrated in operation	

remains open to unclassified methods.

The line between a framework paper and a deployment claim is sharp. This paper satisfies the requirements of the first column of Table 13 and none of the second. The scope of the claims should be read accordingly.

The intelligence framing deserves equal honesty. The argument here is not that lakes are invisible to a determined foreign state; overhead collection and human sources defeat concealment on some timescale, and that timescale is unmeasured. The argument is narrower: the lake changes the surface available to most physical-layer hostile actions, complicates the cheap options, and keeps legal control where it belongs, and whether those changes justify the premium of Section 7 is the open question, not the settled answer.

13 Conclusion

This paper set out to answer a question that had not been isolated in the underwater data center literature: whether

an inland freshwater lake can serve as the physical home of a state's most sensitive computation, and specifically what submersion adds after domestic jurisdiction, remote siting, energy proximity, and multi-site redundancy are held equal. The answer, on the evidence assembled here, is conditional. Inland submersion removes selected direct maritime access paths, notably submarine and surface-vessel interference, and it may offer useful thermal and physical-access trade-offs; it also imports pressure-vessel engineering, an unproven retrieval concept, acoustic exposure through the water column, and a stricter ecological burden. Whether these benefits outweigh those burdens remains unresolved. The framework itself is the contribution: a bounded threat model with published ordinal scores and adversary objectives; a service definition with synchronous replication, witness-based fencing, and both capacity and quorum criteria; a screening of seven Indonesian lakes in which five pass the depth filter, with published inputs, provenance, weight sensitivity, and an explicit demonstration that score error, not weight error, is the larger uncertainty; screening thermal balances whose footprint is reported as a heat-equivalent radius under prescribed detection thresholds; a cost model at equal service in which the decisive comparator, the lake-cooled terrestrial facility, costs 6.34 against 7.85 million dollars for the submerged design; and a documented Monte Carlo model of correlated failure yielding a capacity availability of 0.9952 and a quorum availability of 0.9484, with recovery to the long-run level within four to six months after a regional event.

The next stage should compare matched terrestrial, lake-cooled terrestrial, and submerged options at candidate sites, then test the ranked sites through measured thermal, geotechnical, and operational evidence, exactly the program of Phase 0–3 of Section 11. The deployment program also intersects with existing lake governance. Presidential Regulation 60/2021 established the national priority lakes program and its rescue coordination team (Tim Penyelamatan Danau Prioritas Nasional), covering fifteen lakes, several of which appear in the candidate analysis of this paper [30]. Two implications follow. The first is a constraint: any deployment in a priority lake must be aligned with the rescue and conservation governance, permitting, and monitoring obligations established by that framework, and a security program that bypasses it would generate institutional friction of exactly the kind the program is meant to reduce. The second is an opportunity: the instrumented pilot of Phase 3 produces precisely the data that priority lake rescue requires, bathymetry, thermal profiles, dissolved oxygen, sedimentation rates, and continuous water-quality records; whether a national security program should fund monitoring that conservation budgets struggle to maintain must be compared against the cost of funding the same monitoring without a data center attached. If the pilot violates its thermal budget, or the seismic and pressure tests fail their criteria, the proposal should be withdrawn from consideration; if it does not,

Indonesia gains a candidate class of infrastructure that matches its geography, depth inland, where sovereignty is least negotiable, and the measurements, not this paper, will decide which is the case.

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